your kitchen counter. But crucial information is still hidden within the fuzzy patch. Remember, knowing how things begin—the order of the stack of pages of *War and Peace*, the pressurized carbon dioxide molecules in your bottle of Coke, the state of the universe at the big bang—is essential to understanding how they evolve. Entropy can increase only if it is given room to increase. Entropy can increase only if it starts out low. If the pages of *War and Peace* begin thoroughly jumbled, further tosses will merely leave them jumbled; if the universe started out in a thoroughly disordered, high-entropy state, further cosmic evolution would merely maintain the disorder.

The history illustrated in Figure 9.2 is manifestly not a chronicle of eternal, unchanging disorder. Even though particular symmetries have been lost through cosmic phase transitions, the overall entropy of the universe has steadily increased. In the beginning, therefore, the universe must have been highly ordered. This fact allows us to associate "forward" in time with the direction of increasing entropy, but we still need to figure out an explanation for the incredibly low entropy—the incredibly high state of uniformity—of the newly born universe. This requires that we go even farther back than we have so far and try to understand more of what went on at the beginning—during the fuzzy patch in Figure 9.2—a task to which we now turn.